Credit Card Customer Segmentation: Workflow and Architecture

Overview

This project segments credit card customers into distinct groups so the business can design targeted strategies for different customer types.[file:1] The notebook uses a customer dataset with demographic, relationship, credit, and transaction features, and applies KMeans clustering as the core segmentation method.[file:1]

Dataset

The notebook works with 10,127 customer records and 14 columns.[file:1] The input features include customer demographics such as age, gender, education level, marital status, and dependent count, along with behavioral and financial variables such as estimated income, months on book, relationship count, inactivity, credit limit, transaction amount, transaction count, and utilization ratio.[file:1]

Workflow

The project follows a structured machine learning workflow from raw data inspection to final customer segmentation.[file:1]

1. **Data loading and inspection**: The notebook imports the dataset, previews sample records, checks data types, and confirms that all 14 columns are complete with no missing values.[file:1]
2. **Exploratory data analysis**: It summarizes numerical variables, reviews category distributions for gender, education level, and marital status, and visualizes patterns to understand customer behavior before modeling.[file:1]
3. **Preprocessing**: Categorical variables are encoded, numerical features are transformed and scaled, and the dataset is prepared in a model-ready format suitable for clustering.[file:1]
4. **Clustering with KMeans**: The notebook applies KMeans to group similar customers into clusters based on their combined profile.[file:1]
5. **Cluster selection**: It evaluates candidate cluster counts using the elbow method and silhouette scores; the elbow curve suggests focusing on roughly 4 to 6 clusters, while silhouette scores are computed for multiple values of k to compare separation quality.[file:1]
6. **Dimensionality reduction with PCA**: Because the notebook notes low silhouette scores and correlated features, it introduces PCA to reduce dimensionality and improve clustering interpretability.[file:1]
7. **Cluster assignment and output**: Final cluster labels are attached back to customer records so each customer can be mapped to a segment for downstream business use.[file:1]

Architecture

The project architecture can be understood as a pipeline with five connected layers.[file:1]

1. Data layer

The raw input is a tabular customer dataset containing demographic, account tenure, relationship, transaction, credit, and utilization variables.[file:1]

2. Preparation layer

This layer performs data validation, categorical encoding, feature transformation, and scaling so that features are numerically consistent for distance-based clustering methods such as KMeans.[file:1]

3. Analysis layer

This layer includes descriptive statistics, category frequency checks, and visual exploration to identify distributions, data quality issues, and potential feature relationships before clustering.[file:1]

4. Modeling layer

KMeans is used as the main segmentation algorithm, while elbow and silhouette analysis support cluster-count selection.[file:1] PCA is added as a supporting component to reduce feature redundancy and help address weak cluster separation in the original feature space.[file:1]

5. Output layer

The final output is a customer-level segmentation result in which each customer receives a cluster label.[file:1] These segments can then support differentiated marketing, customer engagement, retention, or credit strategy decisions.[file:1]

Pipeline diagram

```
Raw Customer Data
        |
        v
Data Inspection and EDA
        |
        v
Preprocessing
(encoding + transformation + scaling)
        |
        v
KMeans Clustering
        |
        +----------------------+
        |                      |
        v                      v
```

```
Elbow Method            Silhouette Analysis
        |                      |
        +---------- Model Selection --------+
                                            |
                                            v
                                  PCA-assisted Analysis
                                            |
                                            v
                                Final Cluster Assignment
                                            |
                                            v
                              Customer Segments for Business Use
```

Business use

The notebook is designed to convert raw customer behavior data into actionable segments for strategy design.[file:1] In practice, this means the output can be used to identify high-value customers, low-engagement customers, high-utilization customers, or other behavior-based groups for tailored interventions.[file:1]